

PostgreSQL + Power BI

Product Sales Performance Analysis

Independent Analytics Project

July 2026 - August 2026

Project Overview

This project is an improved and visualized version of my previous SQL sales analysis project, which involved a dataset containing approximately 50,000 sales transactions from a multichannel record of products across different categories and countries. This project extends the previous analysis by transforming important findings from the SQL analysis into interactive visualizations and providing a broader view of sales performance, returns, sales channels, and operational activities.

Objectives

The main objectives of this project are to:

- Determine the monthly sales revenue trend.
- Identify the rankings of products and categories based on different performance metrics.
- Evaluate performance across returns, sales channels, payment methods, shipping, warehouse, and order operations.
- Compare different performance metrics to identify meaningful patterns and differences.
- Construct insights and recommendations based on the results of the analysis.

Tools & Technologies

- PostgreSQL
- Microsoft Power BI
- DAX
- Star Schema Modeling

Dataset

The dataset was sourced from Kaggle and consists of approximately 50,000 sales transactions involving both online and in-store transactions. It contains information



related to products, categories, countries, sales channels, payment methods, shipping providers, warehouses, order priorities, returns, quantities, prices, discounts, and transaction dates.

The dataset is synthetic; therefore, the findings presented in this project are intended to demonstrate the analytical process and methodology rather than represent the actual performance of a real company.

Data Preparation & Modeling

Data Preparation

Before the data could be transformed into actionable insights, it underwent a process of inspection, cleaning, and preparation to improve its consistency and reliability for analysis.

1. **Data Import and Initial Inspection**

The dataset was imported into Power BI and visually inspected for obvious data issues, inconsistencies, and values that could affect the analysis.

2. **Duplicate Inspection and Removal**

Duplicate records were inspected and removed where necessary to prevent repeated transactions from artificially affecting revenue, invoice counts, and other metrics.

3. **Category Standardization**

The original category column was found to contain data-quality issues. It was therefore removed and replaced with a cleaned category column to provide a more consistent classification of products.

4. **Transaction Classification**

Transactions containing negative values were identified and classified separately as negative transactions. This prevented these records from affecting the primary positive-sales analysis while retaining them as an operational and data-quality indicator.

5. Measure-Ready Data Preparation

Relevant date, numerical, categorical, and status fields were prepared for use in Power BI measures, slicers, and visualizations.

Data Model

Fact Sales

The **FactSales** table serves as the central fact table and contains the transaction-level sales records. It stores the measurable business events and numerical information required for the analysis, including invoice information, quantity, unit price, discount, shipping cost, and return status.

Dimension Tables

Dimension tables provide descriptive information that allows the transactions in the fact table to be analyzed from different perspectives.

- **DimDate** — provides year and month information for time-based analysis and monthly revenue trends.
- **DimProduct** — provides product and category information for product and category performance analysis.
- Other descriptive attributes, such as country, sales channel, payment method, shipment provider, warehouse location, order priority, and return status, provide additional analytical context.

Relationships

The fact table is connected to the relevant dimension tables through their corresponding keys. This relationship structure allows selections made through slicers and visual interactions to filter the transaction records and update the relevant measures throughout the dashboard.



For example, selecting a particular country can affect revenue, invoice count, AOV, return metrics, and operational results across the dashboard.

Why the Model Was Important for Filtering and Analysis

The star schema separates the transactional information from the descriptive information used to analyze it. The FactSales table represents **what happened**, while the dimension tables provide context about **what, when, and where the transactions occurred**.

This structure allowed the same measures to be reused across different analytical perspectives. For example, Positive Revenue could be analyzed by product, category, country, sales channel, payment method, shipment provider, warehouse, or order priority without requiring a separate revenue calculation for each visual.

The model therefore provided a reusable foundation for consistent filtering, cross-analysis, and interactive dashboard behavior.

Measures & Analytical Approach

The measures developed for the dashboard were selected based on the business questions the project aimed to answer. Rather than relying only on a single sales metric, the analysis compared revenue, transaction volume, average transaction value, returns, and operational indicators to provide a more complete view of performance.

Sales Performance Measures

Positive Revenue

Positive Revenue measures the revenue generated from valid positive sales transactions while excluding negative transaction values from the primary sales analysis.



This measure was important because negative transactions were classified separately to prevent them from distorting the main sales-performance results.

It answers the question:

How much revenue was generated from the positive sales activity being analyzed?

Total Revenue

Total Revenue provides a broader revenue figure based on the transaction data.

It serves as a reference point that can be compared with Positive Revenue to understand the overall revenue recorded in the dataset and the potential effect of transactions outside the positive-sales subset.

Invoice Count

Invoice Count measures transaction volume.

Revenue alone does not indicate how frequently transactions occurred. A country or product may generate high revenue through a relatively small number of transactions, while another may have a large number of invoices but a lower revenue contribution per transaction.

Invoice Count therefore provides the **volume perspective** of sales performance.

Average Order Value (AOV)

Average Order Value measures the average revenue generated per invoice.

Conceptually:

AOV = Positive Revenue ÷ Invoice Count

Positive Revenue was used as the basis for AOV to maintain consistency with the revenue definition used in the primary sales analysis.



AOV provides a different perspective from total revenue and invoice count. For example, a sales channel may generate greater overall revenue and transaction volume while another channel generates a higher average value per transaction.

Return Analysis

Return Rate

Return Rate measures the proportion of transactions associated with returns.

This metric answers:

How frequently do returns occur?

Return Rate was analyzed separately from returned revenue because a high frequency of returns does not necessarily correspond to the highest financial impact.

Returned Revenue

Returned Revenue measures the revenue associated with returned transactions.

This metric answers:

How much revenue is financially associated with returned sales?

The distinction between Return Rate and Returned Revenue allows the analysis to identify different areas requiring attention. A category may have the highest return rate while another category has the highest returned revenue.

Therefore:

Return Rate = frequency of returns

Returned Revenue = financial impact of returns

Sales Channel Analysis

Sales channels were evaluated using Positive Revenue, Invoice Count, and AOV.

These three measures were analyzed together because each provides a different perspective:

- **Positive Revenue** — overall financial contribution.
- **Invoice Count** — transaction activity.
- **AOV** — average transaction value.

This approach prevents a single metric from being interpreted as the overall indicator of channel performance.

Payment Method Analysis

Payment methods were compared to determine differences in transaction and revenue contribution across the available payment options.

The observed ranking was:

1. Bank Transfer
2. Credit Card
3. PayPal

Although Bank Transfer ranked first, the relatively close results between the payment methods meant that it was not interpreted as overwhelmingly dominant.

Shipping Analysis

Shipping providers were evaluated using invoice count and average shipping cost per invoice.

Invoice Count measures the volume handled by each provider, while Average Shipping Cost per Invoice provides a more comparable measure of the average shipping cost associated with each transaction.



This distinction revealed that the provider handling the greatest volume was not necessarily the provider with the highest average shipping cost.

Warehouse & Order Operations

Warehouse performance was evaluated using Invoice Count and Positive Revenue to compare transaction activity with financial contribution.

Order priorities were analyzed using Invoice Count, Positive Revenue, and AOV. These measures allowed transaction volume, financial contribution, and average transaction value to be compared across different priority levels.

Negative Transactions

Negative transactions were retained as a separate operational and data-quality indicator rather than being included in the primary positive-sales analysis.

A total of **2,489 negative transactions** were identified and verified against the SQL analysis.

Separating these transactions allowed the primary sales metrics to remain focused on positive sales activity while still preserving visibility of potentially problematic records that may warrant further investigation.

Overall Analytical Approach

The overall analytical process followed a structured progression:

Business Question → Select Appropriate Metric → Analyze by Dimension → Compare Results → Identify Patterns or Exceptions → Validate Results → Interpret Findings → Develop Recommendations

This approach was used throughout the project to avoid relying on a single metric or isolated result.

For example, sales channel performance was not evaluated using revenue alone. Revenue, invoice count, and AOV were compared to determine whether differences were related to overall financial contribution, transaction volume, or average transaction value.

The same principle was applied to returns, where Return Rate and Returned Revenue were analyzed separately to distinguish between the frequency and financial impact of returns.

This approach helped transform the dashboard from a collection of visualizations into a structured analytical tool for identifying patterns and potential areas for business attention.

Dashboard Overview

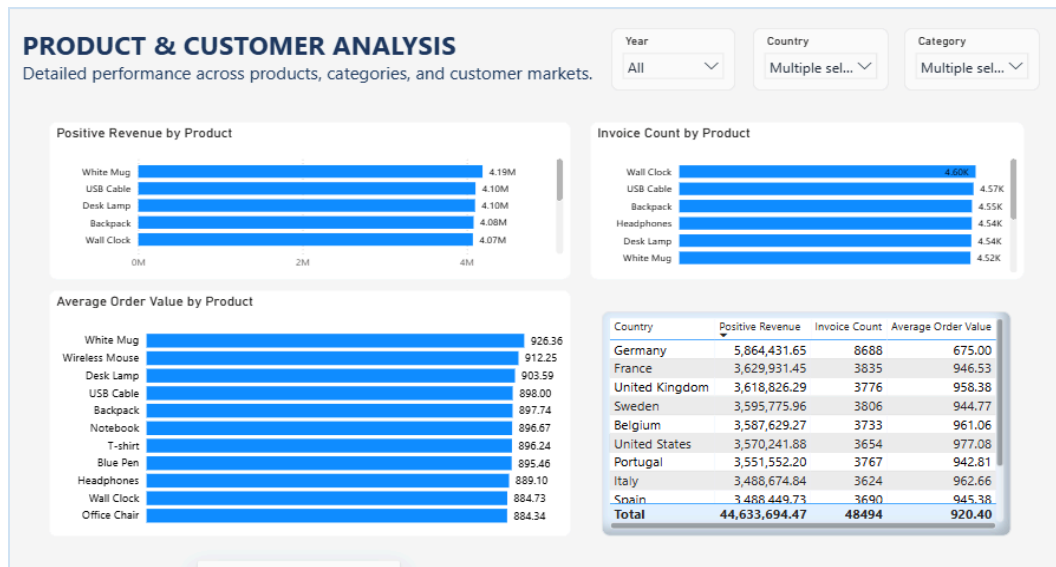
Product Sales Performance

This section presents an overview of sales performance across products, markets, and time. It highlights the main performance indicators, including revenue, invoice count, and average order value.



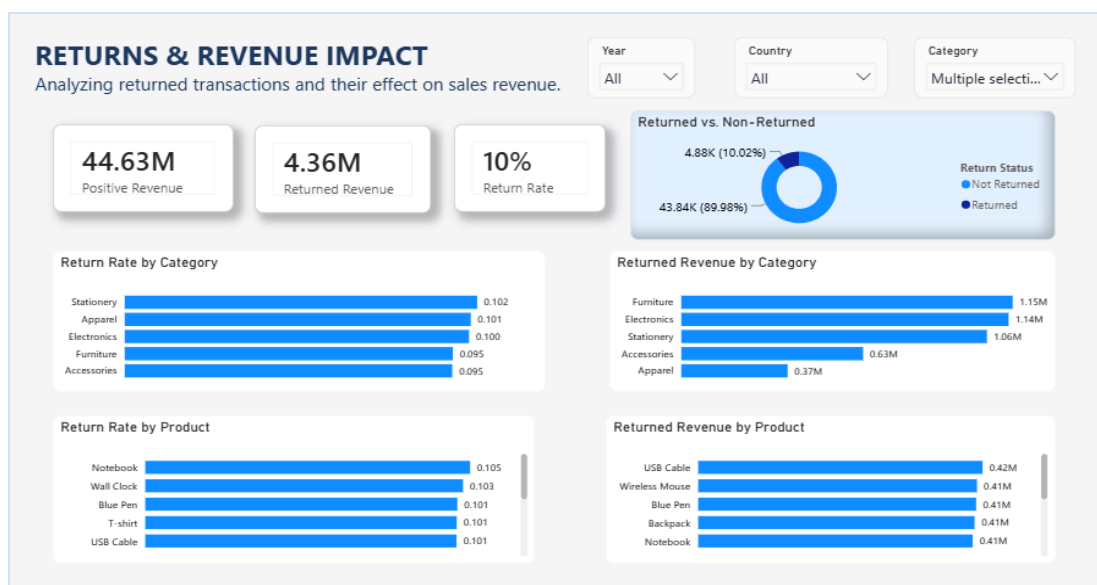
Product & Customer Analysis

This section presents the performance of products, categories, and customer markets. It further investigates how these dimensions rank across revenue, average order value, invoice count, and country.



Return & Revenue Impact

This section analyzes transactions based on return rate and returned revenue. It further determines how products and categories rank in terms of return activity and how returns may potentially affect revenue performance.



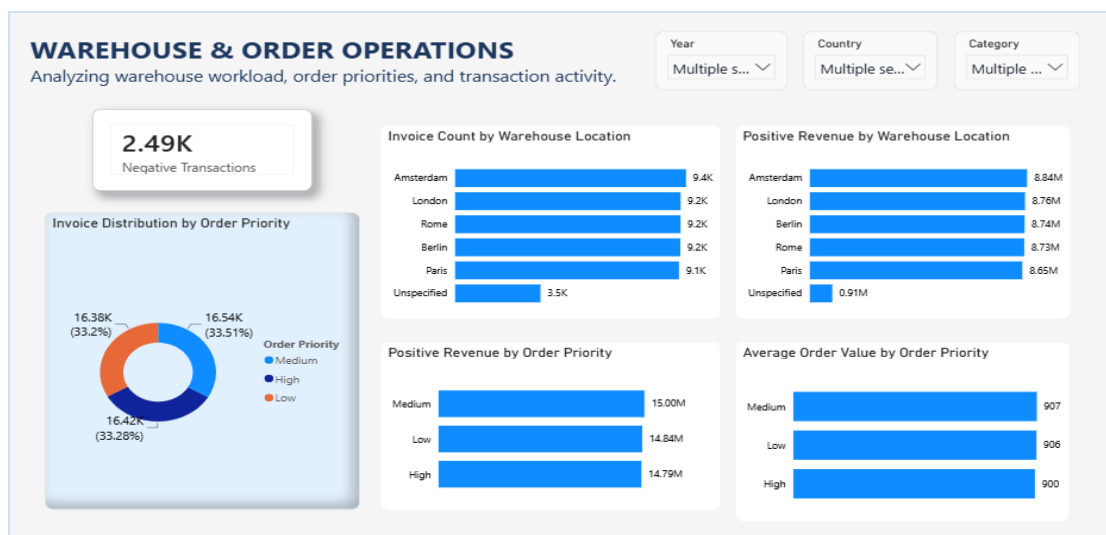
Sales Channel & Operations

This section presents the performance of sales channels, payment methods, and shipping operations using revenue, invoice count, and average order value. These comparisons help assess transaction activity and purchasing patterns across different sales channels and operational methods.



Warehouse & Order Operations

This section analyzes warehouse activity, order priorities, and transaction activity to examine operational patterns across warehouse locations and different levels of order priority.



Key Findings

Across the entire date coverage, customer markets, and product categories, **calculated revenue across all transactions reached 43.51M**, while **positive revenue after excluding negative transactions reached 44.63M**. Positive revenue was used as the baseline for the succeeding analyses to focus on positive sales activity and avoid the effect of identified negative transactions.

The dataset contained **49,782 transaction records**, while the FactSales table contained approximately **48.49K distinct invoices**. The overall average order value was **920.40**.

For category performance, **Electronics** ranked first in positive revenue, while **Apparel** ranked lowest, with category revenue ranging from approximately **12.19M to 4.00M**.

Across customer markets, revenue generated by individual countries ranged from approximately **5.86M to 3.49M**, with **Germany** ranking highest and **Italy** ranking lowest.

Product & Customer Analysis

A more detailed analysis was presented in this section, where product rankings were compared using positive revenue, invoice count, and average order value.

White Mug, USB Cable, and Desk Lamp ranked as the top three products based on positive revenue. For invoice count, **Wall Clock, USB Cable, and Backpack** ranked highest. Meanwhile, **White Mug, Wireless Mouse, and Desk Lamp** ranked highest in average order value.

Among these results, **White Mug ranked first in both positive revenue and average order value**. Overall, the findings demonstrate that product rankings differ depending on whether performance is evaluated through revenue, transaction volume, or average transaction value.

Returns & Revenue Impact

At the category level, **Stationery** ranked first in return rate, while **Furniture** ranked highest in returned revenue. At the product level, **Backpack** ranked first in return rate, while **USB Cable** ranked highest in returned revenue.

These results demonstrate that return rate and returned revenue represent different aspects of return performance. **Return rate measures how frequently transactions are returned, while returned revenue measures the financial value associated with returned transactions.** Therefore, a product or category with a high return rate does not necessarily have the highest financial impact from returns.

Sales Channel & Operations

Online ranked first in both positive revenue and invoice count, while **In-store** ranked highest in average order value. This indicates that Online generated greater sales activity and financial contribution, while In-store transactions had a higher average value per invoice.

For payment methods, **Bank Transfer** ranked first across positive revenue, invoice count, and average order value. However, the results were relatively close across the three payment methods, so the ranking does not necessarily indicate overwhelming dominance.

For shipment providers, **FedEx** ranked first in invoice count and total shipping cost, reflecting its higher transaction volume. Meanwhile, **Royal Mail** ranked highest in average shipping cost per invoice, indicating a higher average shipping cost per transaction despite handling fewer invoices.

Warehouse & Order Operations

A total of **2,489 negative transactions (2.49K)** were identified, representing approximately **5.0% of the 49,782 transaction records**. These transactions were classified separately from positive sales transactions to prevent them from affecting the primary sales-performance analysis while retaining them as an indicator for further data-quality investigation.



For warehouse activity, **Amsterdam** recorded the highest invoice count among the warehouse locations analyzed.

Across order priorities, **Medium** ranked first in invoice count, positive revenue, and average order value. **Low** ranked higher than High in both positive revenue and average order value, showing that transaction volume and transaction value do not necessarily follow the same ranking pattern across order priorities.

Business Recommendations

Based on the observed results, the following areas may warrant further investigation and consideration:

- **Investigate product and category returns** by examining factors that may be associated with higher return activity, such as product characteristics, category, pricing, discounts, or transaction patterns.
- **Examine Royal Mail's average shipping cost per invoice**, which recorded the highest value despite having lower invoice volume than other shipping providers. Further investigation may help determine whether the difference is related to shipment characteristics, destinations, or other operational factors.
- **Evaluate opportunities to increase in-store transaction volume** while maintaining its higher average order value relative to online transactions. This may help determine whether strategies that increase in-store activity can be implemented without reducing transaction value.
- **Strengthen data validation and input controls** to minimize negative transactions and other data-quality issues. Clear validation rules and monitoring procedures could help identify problematic records earlier in the data-collection process.
- **Further investigate the relationship between discounts and other performance metrics**, including revenue, average order value, invoice count, and return activity, to determine whether discount levels are associated with changes in sales performance or returns.

Data Quality and Limitations

- Kaggle-sourced dataset
- Duplicate inspection and removal
- Category data-quality issue and replacement

- **2,489 negative transactions (~5.0% of 49,782 transaction records)**
- Difference between transaction records and distinct invoices
- Assumptions made when defining Positive Revenue, Return Rate, Returned Revenue, and other measures
- Observed relationships indicate patterns or associations but **do not establish causation**
- Findings are intended to demonstrate analytical methodology and should not be interpreted as actual business performance.

Project Structure

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└── Documentation/

Conclusion

Throughout this project, I learned and realized that data analysis involves a rigorous process of handling and preparing data, cleaning and conducting exploratory analysis, managing the desired output, building a data model for visualization, constructing relationships and dimensions, developing visualizations, analyzing results, investigating associations, and constructing insights and recommendations. All of these are not easy tasks, as they require patience, attention to detail, and careful consideration in order to derive meaningful and actionable insights from a previously raw dataset.

This Power BI project is part of the enhancement of my previous Product Sales Analysis project utilizing SQL. The previous project involved extracting data and deriving insights primarily from tabular results, while this project extended the analysis through the



development of an interactive dashboard. The dashboard provides a more comprehensive way of presenting data results and allows users to explore patterns and performance indicators more easily rather than relying only on tables that require the analyst to interpret the results.

In this project, I learned the importance of putting effort and patience into handling data, especially when working with datasets that may contain issues whose causes are not immediately clear. Further, this project helped me understand how an analysis progresses—from inspecting and preparing the data, managing and analyzing it, building a model, visualizing the results in Power BI, and finally interpreting the findings and developing recommendations. The guided practice and online courses I used throughout the process also helped me become more familiar with the different capabilities of Power BI and gave me a better idea of how I can approach my future projects.

This project strengthened my existing analytical foundation and gave me practical experience in applying the data analysis process from preparation to interpretation. I also recognize the value of continued practice and exposure to different datasets and business problems as I further develop these skills. I consider this project a meaningful step in my development as I continue building my data analytics portfolio.

For future projects, I would like to improve my ability to handle and prepare data more efficiently by selecting tools and approaches that are appropriate for the size and complexity of the dataset. I also plan to revisit my existing knowledge of Python and further develop my skills in using it for data analysis.

